

# ZEYU (ALBAN) LI, Ph.D.

*Materials & Bio Scientist-Engineer | Applied AI, Data & Automation for Scientific R&D*

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## SUMMARY

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Cornell Ph.D. in Biological & Environmental Engineering with undergraduate degrees in Chemistry and Computer Science, building AI, data, and automation systems for scientific R&D. Research works across materials science, biomedical engineering, environmental science, and nucleic acid engineering, with a focus on turning complex experimental workflows into structured, reproducible tools. AI and software work includes agent workflows for wet-lab research, hybrid retrieval for researcher discovery, and a hackathon-winning multilingual voice-triage system. Hardware and data work includes microfluidic chip reactors for nucleic-acid production, automated anti-icing test chambers, and reproducible field-campaign data pipelines. Recognized with an outstanding teaching award for AI curriculum design. Record includes 6 publications and 2 granted patents.

## SKILLS

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**LLM Application Stack:** Claude, Gemini, OpenAI APIs; prompt and context engineering; tool/function calling, structured outputs; agent workflow design, multi-agent review; streaming ASR, real-time voice interfaces (WebSocket, Web Audio API); LLM evaluation, hallucination detection; guardrail and safety-layer design, human-in-the-loop validation; AI-assisted development (Claude Code, Cursor, Copilot)

**Machine Learning & Retrieval:** PyTorch, TensorFlow, XGBoost; RAG, semantic search, hybrid retrieval (vector + sparse); query expansion; retrieval evaluation; FAISS, BM25, Reciprocal Rank Fusion; SPECTER2 embeddings

**Data Engineering & Visualization:** Python data stack (pandas, NumPy, GeoPandas, Matplotlib/Seaborn); ETL pipelines, schema validation, multi-source data integration; geospatial analysis, time-series analysis.

**Full-Stack & Deployment:** React 18/19, Electron, Streamlit; Fast API, REST APIs; Firebase (Auth, Firestore, Realtime Database), Vercel; Docker; Git/GitHub

**Experimental Automation Hardware & Scientific Instrumentation:** microfluidics; experimental automation; device fabrication and processing (3D printing: DIW/SLA/FDM, electrospinning, photolithography, thermal imprinting, cleanroom protocols); materials characterization (SEM/EDS, rheometry, DLS, tensile testing, contact-angle measurement); biological assays (qPCR/PCR, in vitro transcription, gel electrophoresis, UV-Vis spectroscopy, fluorometry); design and visualization software (CAD/SketchUp, Blender, KeyShot, Adobe Illustrator/Photoshop)

**Languages:** Python, Java; English, Mandarin, Cantonese

## AI, DATA & HARDWARE AUTOMATION PROJECTS

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### Research-Integrated AI & Scientific Data Systems

**Research Harness** — Agent Skills Framework for Wet-Lab AI-Assisted Research

*Agent Skills, Context Engineering, LLM Reliability, Multi-Agent System, Human-in-the-Loop*

- **Engineered a 13-skill agent framework for LLM-assisted wet-lab research**, translating tacit experimental judgment into reusable agent workflows, including structured pre-output clarification, PASS/FAIL constraint checks, and multi-agent review before final outputs.
- **Built and refined the framework across 130+ AI research sessions over the lifecycle of a wet-lab project**, producing 2,269 indexed artifacts across 167 directories; implemented bidirectional cross-references so future researchers can trace written outputs back to source data and claims; encoded 33 researcher-defined rules and constraints into later agent sessions to better align AI behavior with human wet-lab research judgment.
- **Two workshop papers under review:** ACM CAIS Agent Skills and ICML AI for Science. | GitHub: [github.com/AlbanLi0314/research-harness-skills](https://github.com/AlbanLi0314/research-harness-skills)

**Automated Field Campaign Data Pipeline** — Reproducible Data Engineering for a DoD-Funded Field Study

*Python, Pandas, GeoPandas, Seaborn | ETL, Geospatial Analytics, Data Validation, Scientific Visualization*

- **Built a multi-source ETL pipeline for a DoD SERDP-funded lake-scale field campaign**, integrating ~1,000 qPCR biological measurements and ~1,400 GPS drifter coordinates from 278 sampling stations across an 11 sq. km deployment area;

automated data validation, temporal synchronization, coordinate standardization, and structured multi-table outputs for hydrodynamic model benchmarking.

- **Developed a reproducible figure-generation framework** with ~20 Python scripts and strict schema enforcement, enabling zero-touch regeneration of publication-ready geospatial maps and statistical plots from raw field data

#### **Resonance** — AI-Powered Researcher Discovery Platform

*Deployed on Streamlit | Gemini API, SPECTER2, FAISS, BM25, RAG*

- **Built a paper-grounded researcher discovery platform** that analyzes an uploaded papers, uses a multi-turn Gemini dialogue to clarify the user's discussion intent, and returns ranked researchers with AI-generated relevance explanations.
- **Engineered and evaluated a hybrid academic retrieval pipeline over 230 Cornell researchers**, Gemini-driven query expansion, followed by SPECTER2 dense retrieval via FAISS, BM25 sparse retrieval, and Reciprocal Rank Fusion. Evaluated with 45 human-labeled queries, achieving 77.8% P@1, 0.884 nDCG@5, and 0.847 MRR.

### **Experimental Automation Hardware**

#### **Automated Inline Chip Reactor** — Continuous-Flow Chip Reactor for Nucleic Acid Production

*Microfluidic, Cartridge-Based Reactor, Inline Reactor, Bioprocessing*

- **Designed and assembled modular continuous-flow chip reactors** integrating DNA-immobilized NucleoMesh cartridges, PDMS spacers, acrylic plates, tubing, syringe-pump flow control, and temperature-controlled operation for continuous RNA/DNA production.
- **Built and optimized the reactor hardware workflow** across flow rate, enzyme delivery, temperature, and reagent conditions, supporting multi-day chip operation and long-duration nucleic acid production in a cartridge-based continuous-flow format.

#### **Automated Environment Simulation Chamber** — Droplet-Impact Testing Hardware for Icing Studies

*Environmental Simulation, Experimental Automation, Droplet Control, Electrostatic Surface Charging, High-Speed monitoring*

- **Designed and fabricated an automated icing-environment simulation chamber** for controlled droplet-impact testing, integrating temperature and humidity control, supercooled droplet generation, adjustable droplet (angle, speed, frequency, size), sample-stage cooling and rotation, control panels, and high-speed monitoring.
- **Integrated an electrostatic surface-charging module** with a negative ion generator, high-voltage power supply, grounded shield, and sample stage to simulate charged icing environments and quantify voltage-dependent droplet contact time, expansion ratio, and rebound behavior across superhydrophobic surfaces.

### **AI-Powered Applications**

#### **Kinapse** — Voice-First AI Proficiency Coach

*Gemini Live Agent Hackathon | Multimodal Agents, Real-Time Streaming, RAG, Intent Classification*

- **Built a voice agent that helps users improve their use of LLM tools over time on macOS**, converts natural speech into optimized prompts in real time, displays the rewritten prompt alongside the original for immediate comparison, and tracks per-user prompt-quality progression across sessions.
- **Engineered the real-time agent stack** using Gemini Live streaming ASR with sub-second end-to-end latency, multimodal context fusion across voice, screen vision, and active-window state, and a two-layer RAG memory system combining user profile and episodic history to support personalized coaching and proficiency tracking.

#### **ER Companion** — Multilingual AI Triage Agent for Emergency Departments

*Cornell AI Health Hackathon | Winning Project | Gemini Live API, WebSocket Streaming, Function Calling, Multilingual NLP*

- **Built a multilingual AI voice-triage system for non-English-speaking ER patients**, capturing structured symptom histories in real time across 8 automatically detected languages and delivering parsed intake data to a nurse-facing dashboard; implemented OPQRST-guided questioning through Gemini function calls and dual-layer documentation preserving both the patient's original wording and an AI-generated English paraphrase.
- **Designed the clinical safety layer with dual-track red-flag detection**, 150+ multilingual regex patterns across 4 condition categories for instant frontend alerts, combined with Gemini function-call severity screening using ESI-2 threshold logic and special-population adjustments for elderly, pediatric, and immunocompromised patients. Added automated 7-minute check-ins to track pain-score changes over time. | GitHub: [github.com/AlbanLi0314/er-companion](https://github.com/AlbanLi0314/er-companion)

## RESEARCH EXPERIENCE

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### Graduate Research Assistant

Feb 2021 – May 2026

DNA Materials Lab, Advisor: Prof. Dan Luo | Cornell University, Ithaca, NY

Environmental Monitoring | Molecular Detection | Polymer Engineering

#### DNA-Barcoded PLGA Microparticle Tracers for Environmental DNA Tracing

- **Designed, fabricated, and deployed DNA-barcoded biodegradable PLGA microparticle tracers** in a DoD SERDP-funded lake-scale monitoring study, coordinating a 33-hour, 11 sq. km field campaign across 4 disciplines and 7 research groups.
- **Developed the experimental and analytical workflow** for tracer recovery and TaqMan qPCR detection, enabling quantitative tracking over 7 km from milligram-scale DNA inputs under contamination-controlled conditions.
- **First-author paper** in *Environmental Science & Technology* (IF = 11.3), 2025.

Biomaterials | Self-Healing Materials | Computational Design

#### Biomass DNA Hydrogels for Multi-Mode Crosslinking and Structural Repair

- **Developed a biomass-DNA hydrogel platform** using thermal, photochemical, and metal-ion crosslinking, spanning reversible, photo-cleavable, and irreversible network behaviors from a single low-cost DNA feedstock.
- **Established material processing and performance benchmarks** through rheology, swelling/retention assays, diffusion-limited gelation studies, tensile testing, and extrusion-based 3D printing.
- **Developed DNA hydrogel systems for architectural ceramic tile prototypes**, enabling proof-of-concept self-healing structural repair through separate internal channels for DNA and  $AlCl_3$  solutions that form localized gel at crack sites.
- **First-author paper** in *ACADIA, computational design and digital fabrication premier conference*, 2024.

Biomanufacturing | Microfluidic Reactors | Continuous-Flow Processing

#### NucleoMesh Continuous-Flow Platform for DNA/RNA Production

- **Developed a continuous-flow nucleic acid production platform** using DNA-immobilized NucleoMesh cartridges and syringe-pump-operated chip reactors for multi-day enzymatic RNA/DNA production.
- **Optimized and validated NucleoMesh material performance** through hydrogel formulation, template immobilization, SEM, rheology, fluorescence imaging, gel electrophoresis, and template-loading quantification.
- **Second-author paper** in *Nature Sustainability*, in peer review, 2026.

Biochemical Engineering | Nucleic Acid Extraction | Process Engineering

#### Espresso Machine-Inspired DNA/RNA Extraction Platform

- **Designed an espresso-machine-inspired nucleic acid extraction platform** using a pod-based continuous-flow format to target automatic, high-capacity, and high-throughput processing of nucleic acid samples.
- **Established perfluorocarbons as high-capacity DNA capture media**, with perfluorodecalin reaching  $41.73 \mu\text{g}/\mu\text{L}$  adsorption capacity, roughly 800× higher than silica-based methods and translating to an estimated 209 mg DNA loading per pod.
- **Reduced projected consumables cost** by approximately 91% compared with conventional ethanol precipitation while maintaining over 90% DNA capture efficiency.

### Senior Research Assistant & Undergraduate Research Assistant

Jun 2017 - Jan 2021

Microfabrication & Surface Materials Lab, Supervisor: Prof. Kangning Ren | Hong Kong Baptist University

Surface Engineering | Materials Science | Durable Superwettability

#### Durable Superhydrophobic Polymer Surfaces for Icing Control

- **Engineered and validated all-perfluoropolymer monolithic superhydrophobic surfaces** using high-temperature soft imprinting and thermal fusion to combine controlled 3D microtopology with perfluoropolymer chemistry, maintaining topology-specific superwettability under abrasion, compression, corrosive vapors, UV aging, thermal treatment, and 100,000+ folding cycles
- **Third-author paper** in *The Innovation* (IF = 33.2), 2023; **co-inventor**, U.S. Patent 11,839,998; recipient, HKBU Best Undergraduate Thesis Award.

### Research Exchange Trainee

Jun 2018 - Aug 2018

Molecular Basis of Disease Program | Georgia State University, Atlanta, GA

- **Characterized transcription factor-DNA binding kinetics** (PU.1) via LSPR assays, quantifying sequence-specific binding affinities in real time. Best Poster Presentation Award.

## TEACHING & AI CURRICULUM

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### Co-Instructor & Teaching Assistant

Feb 2021 – Present

Cornell University, Department of Biological and Environmental Engineering

- **Designed and delivered a 6-lecture AI tooling module for engineering students**, covering LLM fundamentals, prompt and context engineering, agentic workflow design, and AI-assisted data processing.
- **Led a 40-student biomaterials AI hackathon** with structured deliverables and oral defense, guiding students to translate AI methods into applied scientific workflow designs.
- **Guest lectured on AI-assisted engineering workflows** in 2 additional undergraduate courses, introducing transformer attention, context engineering, the 40-20-40 rule, and practical agent design patterns.
- **Supported teaching across 7 engineering and bioengineering courses** spanning nucleic acid engineering, biomaterials, molecular and cellular bioengineering, watershed engineering, and engineering professionalism; recipient of the Cornell CALS Outstanding Teaching Assistant Award.
- **Supervised 3 undergraduate researchers** on wet-lab protocols, experimental troubleshooting, and data analysis workflows.

## EDUCATION

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### Cornell University

Ithaca, NY

Biological and Environmental Engineering | Concentration: Materials Science, Biological Engineering, Bioenvironmental Engineering.

**Ph.D.** | May 2026 | **M.S.** | Aug 2024 | **M.Eng.** | May 2020

### Hong Kong Baptist University

Kowloon, Hong Kong

Chemistry (Major) | Computer Science (Minor)

**B.Sc. (Hons)** | Nov 2019

## PUBLICATIONS

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Wang, D., **Li, Z.**, Li, J., Han, Y., Sun, T., Li, F., Liu, P. "A Chip Reactor for Perpetual Nucleic Acid Production and On-chip Information Processing." Nature Sustainability, in peer review, 2026.

**Li, Z.**, Ramón, C. L., Koeberle, A., et al. "Tracing Environmental DNA Transport in Large Lakes with Synthetic DNA Microparticles and Hydrodynamic Modeling." Environmental Science & Technology (IF = 11.3), 2025. [DOI](#)

He, C., **Li, Z. (co-first authors)**, Wang, L. X., et al. "PolyTile 4.0: Self-healing Ceramic Tiles." ACADIA 2024. [DOI](#)

Li, W., Chan, C. W., **Li, Z.**, et al. "All-perfluoropolymer, nonlinear stability-assisted monolithic surface combines topology-specific superwettability with ultradurability." The Innovation (IF = 33.2), 4(2), 100389, 2023. [DOI](#)

### WORKSHOP PAPERS

**Li, Z.** "Apprenticeship Without Proximity: An Agent Skills Architecture for Unwritten Experimental Judgment." Under review at ACM CAIS Workshop on Agent Skills, 2026.

**Li, Z.**, Liu, Z., Yang, X., Luo, D. "What the Records Don't Carry: A Position on Researcher-AI Co-Adaptation in Exploratory Laboratory Research." Under review at ICML Workshop on AI for Science, 2026.

## PATENTS

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Ren, K., Wu, H., Wang, Z., Yao, S., Ong, B., Li, W., **Li, Z.**, Sun, H., & Chan, C.W. "Crack engineering as a new route for the construction of arbitrary hierarchical architectures." US Patent 11,839,998 (granted 2023).

Li, Q., **Li, Z.**, & Lin, Z. "Reactor and method of spiral propulsion biomass continuous thermal cracking." Chinese Patent 201711214139.6 (granted 2017).